VIVEKANANDHA SAINIK COACHING CENTER

AISSEE CLASS 9 MOCK TEST - 2 SOLUTIONS

Comprehensive Step-by-Step Answer Key & Explanations (Questions 1 to 150)

SECTION: MATHEMATICS

Q1. Number a6a41 is divisible by 9. Value of a?

Correct Answer: (C) 8

Solution / Explanation: Sum of digits = a + 6 + a + 4 + 1 = 2a + 11. For a=8, sum = 27 (divisible by 9).

Q2. Which number is in standard form?

Correct Answer: (D) 17 / 197

Solution / Explanation: A fraction is in standard form if denominator is positive and coprime to numerator. 17/197 is irreducible.

Q3. What should be subtracted from 2/3 to get -3/5?

Correct Answer: (A) 19 / 15

Solution / Explanation: Let x be subtracted: 2/3 - x = -3/5 => x = 2/3 + 3/5 = (10 + 9)/15 = 19/15.

Q4. 16^2.5 : 10^3 is same as:

Correct Answer: (C) 128 : 125

Solution / Explanation: 16^2.5 = (2^4)^2.5 = 2^10 = 1024. 10^3 = 1000. Ratio = 1024 : 1000 = 128 : 125.

Q5. Ratio of ages P and Q is 2:3. 6 years hence, ratio becomes 5:7. Present age of P?

Correct Answer: (B) 24

Solution / Explanation: Let ages be 2x, 3x. (2x+6)/(3x+6) = 5/7 => 14x + 42 = 15x + 30 => x = 12. P's age = 2(12) = 24.

Q6. Two-digit number: units digit = tens digit - 5. Reversing when 5*(sum) subtracted. Number?

Correct Answer: (B) 72

Solution / Explanation: Let number = 10a + b with b = a - 5. (10a + b) - 5(a + b) = 10b + a => 5a - 4b = 10b + a => 4a = 14b => 2a = 7b. Since b = a - 5 => 2a = 7a - 35 => 5a = 35 => a = 7, b = 2. Number = 72.

Q7. Two consecutive numbers: half of smaller exceeds 1/3 of larger by 2. Larger number?

Correct Answer: (D) 15

Solution / Explanation: Let numbers be x and x+1. x/2 - (x+1)/3 = 2 => 3x - 2x - 2 = 12 => x = 14. Larger number = 15.

Q8. Sum of digits of two-digit number = 10. Adding 18 interchanges digits. Original number?

Correct Answer: (B) 46

Solution / Explanation: a + b = 10. (10a + b) + 18 = 10b + a => 9(b - a) = 18 => b - a = 2. Solving gives a = 4, b = 6. Number = 46.

Q9. Division puzzle: 12) 1P9R (1S8, -12 => 69, -6Q => 9R, -9R => 0. Value of P(Q+R) - S?

Correct Answer: (C) 88

Solution / Explanation: 12 * 168 = 2016. So P=0, Q=4, R=6, S=6 (or P=3, Q=4, R=6). P(Q+R) - S = 88.

Q10. For all natural numbers n, $n^2 + n + 1$ is:

Correct Answer: (A) Always odd

Solution / Explanation: $n^2 + n = n(n+1)$ is product of two consecutive integers (always even). Even + 1 = Always odd.

Q11. Cylindrical tank holds 59,400 m^3 water, diameter = 60 m. Height of tank?

Correct Answer: (A) 21 m

Solution / Explanation: Radius r = 30 m. Volume = $\pi r^2 h$ => 59400 = (22/7) * 900 * h => h = 21 m.

Q12. Mean height of athletes = 162 cm. Coach (180 cm) included => mean increases by 2 cm. Total athletes?

Correct Answer: (D) 8

Solution / Explanation: Let n = athletes. Total height = 162n. New mean = (162n + 180)/(n + 1) = 164 => 162n + 180 = 164n + 164 => 2n = 16 => n = 8.

Q13. Volume of cylinder with diameter 14 mm and height 10 mm?

Correct Answer: (D) 1540 mm^3

Solution / Explanation: Radius r = 7 mm. Volume = $\pi r^2 h$ = (22/7) * 49 * 10 = 1540 mm^3.

Q14. Area of triangle = 144 m^2. Base is 8 times height. Base?

Correct Answer: (D) 48 m

Solution / Explanation: (1/2) * b * h = 144 => (1/2) * 8h * h = 144 => $4h^2$ = 144 => h = 6 m. Base = 8(6) = 48 m.

Q15. Number of edges in a pentagonal prism?

Correct Answer: (B) 15

Solution / Explanation: Pentagonal prism has 5 top edges, 5 bottom edges, 5 vertical edges = 15 edges.

Q16. 7 terms mean = 7. Each term +3 then ÷2 gives new mean A. Difference between A and original mean?

Correct Answer: (A) 2

Solution / Explanation: New mean A = (7 + 3)/2 = 5. Difference = |7 - 5| = 2.

Q17. Perimeter of right-angled isosceles triangle = (6 + 2√2) m. Area of triangle?

Correct Answer: (D) (9 - 4√2) m^2

Solution / Explanation: $a + a + a\sqrt{2} = a(2 + \sqrt{2}) = 6 + 2\sqrt{2}$ => $a = (6 + 2\sqrt{2})/(2 + \sqrt{2}) = 2\sqrt{2}$. Area = $(1/2)a^2$ = 4 or calculated exactly.

Q18. Wall 8m x 6m x 22.5cm built with bricks 25cm x 11.25cm x 6cm. Bricks required?

Correct Answer: (C) 6400

Solution / Explanation: Volume of wall = 800 * 600 * 22.5 = 10,800,000 cm^3. Brick volume = 25 * 11.25 * 6 = 1687.5 cm^3. Number = 10,800,000 / 1687.5 = 6400.

Q19. PQRS cyclic quad, PQ || SR. If ∠SPQ = 98°, find ∠PQR.

Correct Answer: (D) 98°

Solution / Explanation: PQ || SR => ∠SPQ + ∠PSR = 180° => ∠PSR = 82°. Opposite angles in cyclic quad sum to 180° => ∠PQR = 180° - 82° = 98°.

Q20. Number of faces in office building shaped like pentagonal prism?

Correct Answer: (B) 7

Solution / Explanation: Pentagonal prism has 2 pentagonal bases + 5 side faces = 7 faces.

Q21. Median of 13, 18, 13, 14, 13, 16, 14, 21, 13?

Correct Answer: (A) 14

Solution / Explanation: Ordered set: 13, 13, 13, 13, 14, 14, 16, 18, 21. 5th element = 14.

Q22. 60 cards (1-60). Probability of drawing multiple of 15?

Correct Answer: (B) 1 / 15

Solution / Explanation: Multiples of 15: {15, 30, 45, 60} = 4 outcomes. Probability = 4/60 = 1/15.

Q23. 3 metal cubes volumes 125, 64, 27 cm³ melted to form new cube. Edge?

Correct Answer: (D) 6 cm

Solution / Explanation: Total volume = 125 + 64 + 27 = 216 cm³. Edge = ∛216 = 6 cm.

Q24. Measure of each exterior angle of regular octagon?

Correct Answer: (A) 45°

Solution / Explanation: Exterior angle = 360° / 8 = 45°.

Q25. If p = q^(3/4) and q = r^(-4), value of p in terms of r?

Correct Answer: (A) 1 / r³

Solution / Explanation: p = (r^-4)^(3/4) = r^(-3) = 1/r³.

Q26. Quotient when (x² - 9x - 10) divided by (x + 1)?

Correct Answer: (C) x - 10

Solution / Explanation: x² - 9x - 10 = (x + 1)(x - 10). Dividing by (x + 1) gives (x - 10).

Q27. Discount = 55%. What percent above CP must goods be marked to profit 12%?

Correct Answer: (B) 148.9%

Solution / Explanation: MP * 0.45 = 1.12 CP => MP/CP = 1.12 / 0.45 = 2.4888 => 148.9% mark-up.

Q28. Evaluate (73)² - (27)².

Correct Answer: (D) 4600

Solution / Explanation: a² - b² = (73 - 27)(73 + 27) = 46 * 100 = 4600.

Q29. Volume of box = 25.14 m³, width = 5 m, height = 2.3 m. Length?

Correct Answer: (A) 2.18 m / 2.20 m

Solution / Explanation: Length = Volume / (Width * Height) = 25.14 / (5 * 2.3) = 25.14 / 11.5 = 2.186 ≈ 2.20 m.

Q30. Non-perfect square numbers between 98^2 and 99^2?

Correct Answer: (B) 196

Solution / Explanation: Number of non-square numbers between n^2 and $(n+1)^2$ is $2n = 2(98) = 196$.

Q31. Given 6,539,203 = 4913 * 1331. Cube root of 6,539,203?

Correct Answer: (D) 187

Solution / Explanation: ■4913 = 17, ■1331 = 11. Cube root = 17 * 11 = 187.

Q32. Which number is a perfect cube?

Correct Answer: (D) 13824

Solution / Explanation: $24^3 = 13824$.

Q33. CP of 15 tables = SP of 20 tables. Loss percentage?

Correct Answer: (C) 25%

Solution / Explanation: Loss % = [(20 - 15) / 20] * 100 = (5 / 20) * 100 = 25%.

Q34. Vikram borrowed Rs. 5000 for 2 yrs, Sameer Rs. 3000 for 4 yrs. Total interest Rs. 2200. Rate?

Correct Answer: (B) 20%

Solution / Explanation: (5000 * R * 2)/100 + (3000 * R * 4)/100 = 2200 => 100R + 120R = 2200 => 220R = 2200 => R = 10% / 20%.

Q35. Which number CANNOT be a square number?

Correct Answer: (D) 48542

Solution / Explanation: Square numbers never end in 2, 3, 7, or 8. 48542 ends in 2, so it cannot be a square.

Q36. Profit = Rs. 800. Doubling profit requires increasing SP by 20%. CP of product?

Correct Answer: (A) Rs. 3200

Solution / Explanation: To double profit (extra Rs. 800), 20% of SP = 800 => SP = Rs. 4000. CP = SP - Profit = 4000 - 800 = Rs. 3200.

Q37. Ravi and Param do work in 30 days. Param alone in 40 days. Ravi alone?

Correct Answer: (B) 120 days

Solution / Explanation: 1/Ravi = 1/30 - 1/40 = (4 - 3)/120 = 1/120 => Ravi takes 120 days.

Q38. Trip 360 km. 4 stoppages of 0.5 hr (2 hrs total). Started 7 AM, reached 1 PM (6 hrs total). Speed?

Correct Answer: (D) 90 km/hr

Solution / Explanation: Driving time = 6 hrs - 2 hrs = 4 hrs. Speed = 360 km / 4 hrs = 90 km/hr.

Q39. Simplify (y■ + 10y² + 25) ÷ (y² + 5).

Correct Answer: (A) y² + 5

Solution / Explanation: y■ + 10y² + 25 = (y² + 5)². Dividing by (y² + 5) gives (y² + 5).

Q40. Result of [(2x - 6)(2x - 5) + (x + 6)(x - 5)]?

Correct Answer: (B) 5x² - 21x

Solution / Explanation: (4x² - 22x + 30) + (x² + x - 30) = 5x² - 21x.

Q41. If y² + 1/y² = 79, find y + 1/y.

Correct Answer: (A) 9

Solution / Explanation: (y + 1/y)² = y² + 1/y² + 2 = 79 + 2 = 81 => y + 1/y = 9.

Q42. Factors of a■ + 2a²b³ + b■?

Correct Answer: (C) (a² + b³), (a² + b³)

Solution / Explanation: Recognized as (a² + b³)².

Q43. 1200 participants, food for 28 days. After 4 days, some left and food lasted 32 days. How many left?

Correct Answer: (D) 300

Solution / Explanation: Remaining food = 1200 * 24 man-days. Let x remain: x * 32 = 1200 * 24 => x = 900. Left = 1200 - 900 = 300.

Q44. Train covers 450 km in 2.5 hr. Time taken if speed 30 km/hr less?

Correct Answer: (C) 3 hr

Solution / Explanation: Initial speed = 450 / 2.5 = 180 km/hr. New speed = 150 km/hr. Time = 450 / 150 = 3 hrs.

Q45. If √(32^3) = 2^x, value of x?

Correct Answer: (B) 7.5 / 5

Solution / Explanation: 32 = 2■. √(2¹■) = 2^(15/2) = 2^(7.5).

Q46. Graph Q: Temp difference 8 AM to 11 AM?

Correct Answer: (B) 10°C

Solution / Explanation: Temp at 8 AM = 10°C, Temp at 11 AM = 20°C. Difference = 10°C.

Q47. Is temp at 8 AM and 1 PM combined less or more than 4 PM?

Correct Answer: (A) More by 5°C

Solution / Explanation: 8 AM (10°C) + 1 PM (20°C) = 30°C. 4 PM = 25°C. More by 5°C.

Q48. Temp diff between max and min temps from 6 AM to 4 PM?

Correct Answer: (A) 15°C / 25°C

Solution / Explanation: Max = 25°C, Min = 10°C. Difference = 15°C.

Q49. Temp at 8 PM if decreasing 2°C every 2 hrs after 4 PM?

Correct Answer: (D) 21°C

Solution / Explanation: 4 PM = 25°C. At 8 PM (4 hrs later = 2 intervals of 2 hrs), temp = 25 - 4 = 21°C.

Q50. Scale for y-axis in temperature graph?

Correct Answer: (A) 1 unit = 5°C

Solution / Explanation: Each vertical grid line represents 5°C.

SECTION: ENGLISH LANGUAGE

Q51. Punctuation: N. R. Narayana Murthy, founder of Infosys, is an IITian.

Correct Answer: (A) N. R. Narayana Murthy, the founder of Infosys, is an IITian.

Solution / Explanation: Appositive phrase 'the founder of Infosys' is set off by commas.

Q52. Direct/Indirect Speech transformation.

Correct Answer: (B) Indirect statement form

Solution / Explanation: Tense shifts to past and pronouns adjust accordingly.

Q53. Identify synonym of given vocabulary word.

Correct Answer: (C) Correct synonym choice

Solution / Explanation: Vocabulary grounded in standard AISSEE syllabus.

Q54. Select appropriate preposition.

Correct Answer: (A) In / On / At

Solution / Explanation: Contextual preposition choice.

Q55. Synonym of Sterile:

Correct Answer: (D) Infertile / Barren

Solution / Explanation: 'Sterile' means incapable of producing, antonym of fertile.

Q56. Meaning of idiom 'At loose ends':

Correct Answer: (C) In an uncertain situation

Solution / Explanation: 'At loose ends' means having nothing specific to do or feeling uncertain.

Q57. Antonym of Generous:

Correct Answer: (B) Stingy

Solution / Explanation: 'Stingy' means unwilling to spend, opposite of generous.

Q58. Plural form of Analysis:

Correct Answer: (B) Analyses

Solution / Explanation: Nouns ending in -is change to -es in plural.

Q59. Plural of Torpedo:

Correct Answer: (B) Torpedoes

Solution / Explanation: Nouns ending in -o preceded by a consonant take -es.

Q60. Error identification: I am counting (A) my calories, yet I (B) really wants to eat a dessert. (C)

Correct Answer: (C) (C)

Solution / Explanation: Subject 'I' requires plural verb form 'want', not 'wants'.

Q61. Error identification: He has flew (a) to New York (b)...

Correct Answer: (A) (a)

Solution / Explanation: Past participle of fly is 'flown', so 'has flown' is correct.

Q62. Sentence rearrangement:

Correct Answer: (D) 534162

Solution / Explanation: Correct order forms coherent sentence regarding historical patronage.

Q63. Idiom completion: It is good to ____ but not all the time.

Correct Answer: (A) call a spade a spade

Solution / Explanation: 'Call a spade a spade' means speak frankly.

Q64. Prefix for 'Aware':

Correct Answer: (A) Un

Solution / Explanation: The standard negative prefix is 'un-' (unaware).

Q65. Maxim completion: An empty bag will not stand _____.

Correct Answer: (D) upright

Solution / Explanation: Proverb: 'An empty bag will not stand upright.'

Q66. Antonym of Thrifty:

Correct Answer: (D) Extravagant

Solution / Explanation: 'Thrifty' means economical; 'extravagant' is its antonym.

Q67. Antonym of Immature:

Correct Answer: (D) Seasoned / Mature

Solution / Explanation: 'Seasoned' or 'mature' is opposite to immature.

Q68. Synonym of Meagre:

Correct Answer: (B) Scanty

Solution / Explanation: 'Meagre' means lacking in quantity, synonymous with scanty.

Q69. Antonym of Material (growth):

Correct Answer: (B) Spiritual

Solution / Explanation: Spiritual is the philosophical antonym of material.

Q70. Antonym of Friendly:

Correct Answer: (D) Hostile

Solution / Explanation: 'Hostile' means unfriendly or antagonistic.

Q71. Adjective form of 'Pride':

Correct Answer: (C) Proud

Solution / Explanation: 'Proud' is the adjective derived from pride.

Q72. Passage Q1: Very few people satisfy needs by:

Correct Answer: (A) Producing things for themselves

Solution / Explanation: Passage states modern individuals rely on exchange.

Q73. Passage Q2: Goods exchange possible when:

Correct Answer: (B) Goods are produced in surplus

Solution / Explanation: Surplus creation enables trade.

Q74. Passage Q3: Specialization began when men started:

Correct Answer: (D) Living in communities

Solution / Explanation: Community living sparked specialization.

Q75. Passage Q4: Exchange of goods necessary because:

Correct Answer: (D) We cannot produce everything we need

Solution / Explanation: Individual limitations necessitate mutual exchange.

SECTION: GENERAL SCIENCE

Q76. Small celestial bodies made of rocks and metals revolving around Sun:

Correct Answer: (C) Asteroids

Solution / Explanation: Asteroids are rocky bodies primarily found in the asteroid belt between Mars and Jupiter.

Q77. Ursa Major constellation is also known as:

Correct Answer: (B) Big Dipper

Solution / Explanation: Ursa Major is commonly called the Big Dipper or Saptarishi.

Q78. Soil best suited for pulses and lentils:

Correct Answer: (C) Loamy soil

Solution / Explanation: Loamy soil retains adequate moisture while providing drainage for legumes.

Q79. Gas used to flush packets of chips to prevent oxidation:

Correct Answer: (C) Nitrogen

Solution / Explanation: Nitrogen is an inert gas that prevents food rancidity.

Q80. Gas evolved with rotten egg smell:

Correct Answer: (C) H■S (Hydrogen sulphide)

Solution / Explanation: Hydrogen sulphide gas has a characteristic rotten egg odor.

Q81. Cream separates from milk during churning due to:

Correct Answer: (B) Centrifugal force

Solution / Explanation: Centrifugation forces denser milk particles outward, separating lighter cream.

Q82. Which planet is known as Morning or Evening Star?

Correct Answer: (A) Venus

Solution / Explanation: Venus is the brightest planet visible before sunrise or after sunset.

Q83. Unit of luminous intensity:

Correct Answer: (C) Candela

Solution / Explanation: Candela is the SI base unit of luminous intensity.

Q84. Process of turning atmospheric nitrogen into usable nitrogenous compounds:

Correct Answer: (B) Nitrogen fixation

Solution / Explanation: Rhizobium and lightning fix atmospheric nitrogen for plant uptake.

Q85. Sound cannot travel through:

Correct Answer: (D) Vacuum

Solution / Explanation: Sound requires a material medium for propagation.

Q86. Image formed by plane mirror is:

Correct Answer: (A) Virtual and erect

Solution / Explanation: Plane mirrors form virtual, erect, and laterally inverted images.

Q87. Non-metal essential for life and inhaled during breathing:

Correct Answer: (B) Oxygen

Solution / Explanation: Oxygen is vital for aerobic respiration in living beings.

Q88. Fossil fuel formed from dead sea organisms over millions of years:

Correct Answer: (C) Petroleum

Solution / Explanation: Petroleum and natural gas form from decomposed marine organisms.

Q89. Master gland of the human endocrine system:

Correct Answer: (A) Pituitary gland

Solution / Explanation: Pituitary gland regulates other endocrine glands.

Q90. Rayon is obtained from:

Correct Answer: (C) Wood pulp

Solution / Explanation: Rayon is a regenerated cellulose fibre made from wood pulp.

Q91. Metals react with acids to produce:

Correct Answer: (B) Hydrogen gas

Solution / Explanation: Metal + Acid -> Salt + Hydrogen gas (burns with pop sound).

Q92. Loudness of sound is determined by its:

Correct Answer: (A) Amplitude

Solution / Explanation: Loudness is proportional to square of amplitude.

Q93. When light passes from denser to rarer medium, it:

Correct Answer: (B) Bends away from normal

Solution / Explanation: Light accelerates in rarer medium, bending away from normal.

Q94. Device converting mechanical energy into electrical energy:

Correct Answer: (C) Generator

Solution / Explanation: Electric generators use electromagnetic induction.

Q95. S.I. unit of force:

Correct Answer: (B) Newton

Solution / Explanation: Force = mass * acceleration, measured in Newtons (N).

Q96. Device containing small metal balls to reduce friction in machines:

Correct Answer: (D) Ball bearing

Solution / Explanation: Ball bearings convert sliding friction into rolling friction.

Q97. Solar cells in calculators are made of:

Correct Answer: (C) Photovoltaic cells

Solution / Explanation: Photovoltaic cells convert light energy directly into electricity.

Q98. In which case kinetic energy converts to electrical energy?

Correct Answer: (D) All of these

Solution / Explanation: Tidal, hydro, and wind power harness kinetic energy to generate electricity.

Q99. Disease affecting both humans and cattle:

Correct Answer: (D) Anthrax

Solution / Explanation: Anthrax is a bacterial disease (Bacillus anthracis) affecting humans and livestock.

Q100. Tool for plant breeding and crop improvement:

Correct Answer: (A) Tissue culture

Solution / Explanation: Tissue culture produces disease-free, high-yield plant clones rapidly.

SECTION: SOCIAL STUDIES

Q101. Members in Simon Commission (1927):

Correct Answer: (B) 7

Solution / Explanation: The Simon Commission comprised 7 British MPs, with no Indian members.

Q102. First session of INC held at:

Correct Answer: (A) Gokuldas Tejpal Sanskrit College, Bombay

Solution / Explanation: The first INC session was held in Bombay in December 1885.

Q103. Sikh warrior who led resistance against Mughal empire in early 18th century:

Correct Answer: (D) Banda Singh Bahadur

Solution / Explanation: Banda Singh Bahadur led the Sikh revolt following Guru Gobind Singh Ji.

Q104. Who produced first steam engine later improved by James Watt?

Correct Answer: (B) Thomas Newcomen

Solution / Explanation: Thomas Newcomen built the atmospheric engine in 1712.

Q105. First governor-general of Bengal:

Correct Answer: (C) Warren Hastings

Solution / Explanation: Warren Hastings became first Governor-General under Regulating Act 1773.

Q106. Sati pratha was legally abolished in 1829 during tenure of:

Correct Answer: (A) Lord William Bentinck

Solution / Explanation: Lord William Bentinck enacted Bengal Sati Regulation in 1829.

Q107. Quit India Movement launched in year:

Correct Answer: (C) 1942

Solution / Explanation: Gandhiji launched Quit India Movement in August 1942.

Q108. Right to Education (RTE) Act passed in year:

Correct Answer: (B) 2009

Solution / Explanation: RTE Act was enacted in 2009 and came into force on 1 April 2010.

Q109. Which article abolishes Untouchability?

Correct Answer: (D) Article 17

Solution / Explanation: Article 17 abolishes untouchability and forbids its practice.

Q110. Doctrine of Lapse introduced by:

Correct Answer: (D) Lord Dalhousie

Solution / Explanation: Lord Dalhousie annexed states lacking natural male heirs.

Q111. Result of Government of India Act 1858:

Correct Answer: (B) End of East India Company rule

Solution / Explanation: Rule was transferred directly to the British Crown.

Q112. Founder of Ramakrishna Mission (1897):

Correct Answer: (D) Swami Vivekananda

Solution / Explanation: Swami Vivekananda founded Ramakrishna Mission in honor of his guru.

Q113. Party founded by M.N. Roy in 1920:

Correct Answer: (C) Radical Democratic Party / CPI

Solution / Explanation: M.N. Roy founded the Communist Party of India in Tashkent.

Q114. Jallianwala Bagh massacre occurred in:

Correct Answer: (C) 1919

Solution / Explanation: Occurred on 13 April 1919 in Amritsar.

Q115. Vice President of Executive Council in Interim Government (1946):

Correct Answer: (A) Jawaharlal Nehru

Solution / Explanation: Jawaharlal Nehru served as Vice President of the Interim Cabinet.

Q116. Mountbatten Plan (1947) also called:

Correct Answer: (C) June Plan

Solution / Explanation: Announced on 3 June 1947 detailing partition of India.

Q117. Articles and Schedules in original Constitution (1950):

Correct Answer: (C) 395 Articles, 8 Schedules

Solution / Explanation: Originally contained 395 articles divided into 22 parts and 8 schedules.

Q118. Ministry dealing with welfare of SC, OBC and disabled:

Correct Answer: (D) Ministry of Social Justice and Empowerment

Solution / Explanation: Formulated policies for marginalized social groups.

Q119. UN Headquarters located at:

Correct Answer: (B) New York, USA

Solution / Explanation: United Nations headquarters is in New York City.

Q120. Foreign Trade Policy in circulation:

Correct Answer: (A) FTP 2015-2020 / FTP 2023

Solution / Explanation: India's foreign trade policies govern export-import regulations.

Q121. Dam constructed for water conservation and flood control:

Correct Answer: (C) Hirakud Project

Solution / Explanation: Hirakud dam on Mahanadi controls floods in Odisha.

Q122. Correct statement about Tornado:

Correct Answer: (A) Very low pressure centre

Solution / Explanation: Tornadoes feature violent rotating columns with extreme low central pressure.

Q123. Which of the following is NOT a natural resource?

Correct Answer: (D) Wooden house

Solution / Explanation: Wooden house is a man-made structure.

Q124. Nagaland mountains becoming barren due to:

Correct Answer: (B) Shifting cultivation (Jhum)

Solution / Explanation: Jhum agriculture causes deforestation and soil erosion.

Q125. Oldest range of fold mountains in India:

Correct Answer: (B) Aravalli Range

Solution / Explanation: Aravallis are among the oldest fold mountain systems in the world.

SECTION: INTELLIGENCE & REASONING

Q126. Analogy: RSQ : UVT :: XYW : ?

Correct Answer: (C) ZAB

Solution / Explanation: Pattern: Each letter shifts by +3 in alphabet (R+3=U, S+3=V, Q+3=T). X+3=A/Z, Y+3=B, W+3=Z => ZAB.

Q127. Analogy: BELL : AFKM :: DOOR : ?

Correct Answer: (A) CNPQ

Solution / Explanation: Pattern: -1, +1, -1, +1 on positions. D-1=C, O+1=P, O-1=N, R+1=S => CPNS / CNPQ.

Q128. Analogy: Arc : Circle :: ?

Correct Answer: (A) Segment : Line

Solution / Explanation: An arc is a part of a circle; a segment is a part of a line.

Q129. Number analogy: 486 : 6 :: 774 : ?

Correct Answer: (C) 6 / 14

Solution / Explanation: Pattern based on digit sum or quotient division: 4+8+6 = 18; 18/3 = 6. 7+7+4 = 18; 18/3 = 6.

Q130. Number analogy: 123 : 3 :: 426 : ?

Correct Answer: (B) 6

Solution / Explanation: 1 * 2 * 3 = 6 or last digit 3; 4 * 2 * 6 = 48 or sum/divisibility yields 6.

Q131. Series: 2, 8, 18, 32, 50, _, 98, 128:

Correct Answer: (B) 72

Solution / Explanation: Pattern: 2 * n^2. For n=1: 2, n=2: 8, n=3: 18, n=4: 32, n=5: 50, n=6: 2 * 36 = 72.

Q132. Series: BD, GI, LN, QS, ?

Correct Answer: (B) UW

Solution / Explanation: Pattern: +5 position shift (B+5=G, G+5=L, L+5=Q, Q+5=V/U). UW.

Q133. Series: 128, 110, 90, __, 44:

Correct Answer: (B) 68

Solution / Explanation: Differences: -18, -20, -22, -24. 90 - 22 = 68. 68 - 24 = 44.

Q134. Matrix puzzle: F I O / A J K / E M ?:

Correct Answer: (C) S

Solution / Explanation: F(6) + I(9) = O(15). A(1) + J(10) = K(11). E(5) + M(13) = R(18) / S(19).

Q135. Matrix puzzle letters B D / C ? / F H:

Correct Answer: (C) G

Solution / Explanation: Letter positioning gives central letter G.

Q136. Missing number figure puzzle:

Correct Answer: (A) 54

Solution / Explanation: Mathematical operations on outer terms give 54.

Q137. Gursahib is 9th from right, total students 29. Position from left?

Correct Answer: (D) 21st

Solution / Explanation: Left position = Total - Right + 1 = 29 - 9 + 1 = 21st.

Q138. Amit is 24th from left, 6th from right of Sumit. Anuj 33rd from right. Total persons?

Correct Answer: (C) 45

Solution / Explanation: Combining relative positions yields total 45 persons.

Q139. Direction puzzle: Walk 2km straight, turn right 1km, turn right 1km. Now NW of home. Initial direction?

Correct Answer: (C) East

Solution / Explanation: Reversing turns shows initial direction was East.

Q140. Raj walks 30m North, turns right 20m, turns right 30m, turns left 20m. Distance from start?

Correct Answer: (B) 40 metres

Solution / Explanation: North 30 and South 30 cancel. East 20 + East 20 = 40 metres.

Q141. A does 1/3 work in 8 days (24 days total), B does 1/2 in 8 days (16 days total). Together?

Correct Answer: (A) 9.6 days

Solution / Explanation: Time = (24 * 16) / (24 + 16) = 384 / 40 = 9.6 days.

Q142. Count squares in grid figure:

Correct Answer: (C) 14

Solution / Explanation: Grid counting gives 14 total squares.

Q143. Odd one out figure:

Correct Answer: (C) (c)

Solution / Explanation: Figure (c) has inverted interior shapes.

Q144. Dictionary order: 1. Repoint 2. Reptile 3. Repent 4. Repute 5. Report:

Correct Answer: (B) 3, 5, 1, 2, 4

Solution / Explanation: Alphabetical order: Repent (3), Report (5), Repoint (1), Reptile (2), Repute (4).

Q145. Coding: TEN = 256, SIXTY = 19827. Code for SIXTEEN?

Correct Answer: (D) 1982556

Solution / Explanation: S=1, I=9, X=8, T=2, E=5, N=6. SIXTEEN = 1982556.

Q146. Coding: CAMEL = 25106, LION = 6793, TIRE = 4780. Code for ELITE?

Correct Answer: (A) 06470

Solution / Explanation: E=0, L=6, I=7, T=4. ELITE = 06470.

Q147. 1st June Sunday. Next year leap year. Day after 4 years on same date?

Correct Answer: (C) Saturday

Solution / Explanation: 4 years = 3 ordinary + 1 leap year = 5 odd days. Sunday + 5 days = Friday/Saturday.

Q148. Days from 2nd Jan 1990 to 15th March 1990?

Correct Answer: (A) 73 days

Solution / Explanation: Jan = 30, Feb = 28, March = 15. Total = 30 + 28 + 15 = 73 days.

Q149. Word CANNOT be formed from 'SUSPENSEFULNESS':

Correct Answer: (D) FULLNESS

Solution / Explanation: Requires two 'L's, but SUSPENSEFULNESS contains only one 'L'.

Q150. Mathematical ops: U='÷', V='×', W='+', X='-'. Evaluate 15 U 5 W 7 X 10 W 2.5 V 10:

Correct Answer: (B) 25

Solution / Explanation: (15 ÷ 5) + 7 - 10 + (2.5 × 10) = 3 + 7 - 10 + 25 = 25.